

PAIRSELECT: TRAINING-FREE TOKEN SELECTION FOR EFFICIENT WEAKLY SUPERVISED VIDEO ANOMALY DETECTION

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ABSTRACT

Detecting rare events requires sustained observation, but not necessarily full computation for every internal video token. We introduce PairSelect, a training-free token selection operator for frozen weakly supervised video anomaly detectors. After dense contextualization in the first half of a video encoder, PairSelect allocates local quotas and retains activation-magnitude-guided representatives without averaging their vectors. It uses adjacent spatial pairs for joint-attention encoders and complete spatial trajectories for divided space–time attention. Neither input clip coverage nor the dense-trained detection head is changed. Across three encoders and two datasets, the completed middle-budget evaluation removes approximately 40–43% of subsequent tokens, with primary-metric changes from -0.16 to $+0.68$ percentage points. Analytical accounting predicts 21.75–23.04% fewer operations in the twelve principal transformer blocks. Additional budgets and measurements of full-encoder FLOPs, latency, throughput, and peak GPU memory remain to be completed.

Index Terms— video anomaly detection, weak supervision, token selection, training-free compression, efficient inference
Author working draft: unmeasured results are marked with dashes; author revision and verification are required before submission.

1. INTRODUCTION

Video anomaly detection faces an asymmetric computational problem: unusual events are scarce, yet a system must process long stretches of ordinary activity to find them. In weakly supervised video anomaly detection (WSVAD), video-level training labels do not identify the anomalous snippets, which can be overwhelmed by normal instances [1, 2]. A practical pipeline extracts clip representations with a pretrained encoder and passes the resulting sequence to a temporal detector [3]. Reducing the number of input clips is one way to lower cost, but it also changes the temporal observations available to that detector. We instead ask how to reduce internal computation while keeping the original clip schedule intact.

The rarity of anomalous events does not imply that every internal token carries unique information. VideoMAE demonstrates the usefulness of video redundancy for masked pretraining [4]; token sampling and merging also exploit redundant representations in pretrained transformers [5, 6]. These findings motivate a different allocation of inference effort: allow every token to participate in the first half of the encoder, then continue processing only a structured subset. Pretraining redundancy alone does not establish that a frozen anomaly detector tolerates this intervention. The resulting representations must still work with its existing detection head.

We introduce PairSelect, a local, non-mixing selection operator for this setting. It combines three choices: spatially adjacent candidate units, fixed group budgets, and an activation-magnitude preference within those units. For joint-attention encoders, a first selection round gives each pair a representative whenever the budget permits; remaining slots retain additional members of stronger pairs. For divided space–time attention, entire spatial trajectories are selected together. The selected vectors are gathered without averaging at the insertion point. No selector is trained, no compression-specific labels are required, and neither the encoder nor the dense-trained head is fine-tuned after insertion.

Our contributions are a training-free, structure-aware selection rule for frozen WSVAD pipelines; explicit budget and computational-cost accounting for joint and divided attention; and a direct-insertion evaluation on three encoders and two datasets. At the completed middle budget, the largest observed primary-metric decrease is 0.16 percentage points, while approximately 40–43% of tokens are removed from the remaining encoder blocks. The budget ladder separates local duplicate removal from more aggressive loss of local representatives, providing a concrete basis for studying quality–cost trade-offs rather than assuming a preferred compression ratio.

2. RELATED WORK

Weakly supervised anomaly detection. Multiple-instance ranking [1], robust temporal feature magnitudes [2], and uncertainty-regulated dual memories [3] address video-level supervision. VadCLIP adapts frozen vision–language representations through a task-specific detection architecture [7]. Efficient anomaly detectors also include learned autoencoder designs [8]. We use UR-DMU as a fixed downstream head within each encoder–dataset configuration, and reduce visual encoding cost rather than introduce another detector. Our setup is visual-only even on the audiovisual XD-Violence benchmark [9].

Efficient transformers. VideoMAE and VideoMAEv2 use masked video pretraining [4, 10]; TimeSformer separates temporal and spatial attention [11]. DynamicViT learns token-importance predictors [12], and EViT reorganizes attentive and inattentive tokens [13]. EVAD studies task-aware token dropout for action detection [14]. ATS provides parameter-free sampling, and ToMe supports merging without additional training [5, 6]. Thus, training-free reduction itself is not new. PairSelect studies a local non-mixing rule at a fixed intermediate depth with an unchanged anomaly head and unchanged clip coverage. Recent ESOM and LAVIDA also incorporate token compression within multimodal language-model pipelines [15, 16]; that setting differs from internal visual-encoder

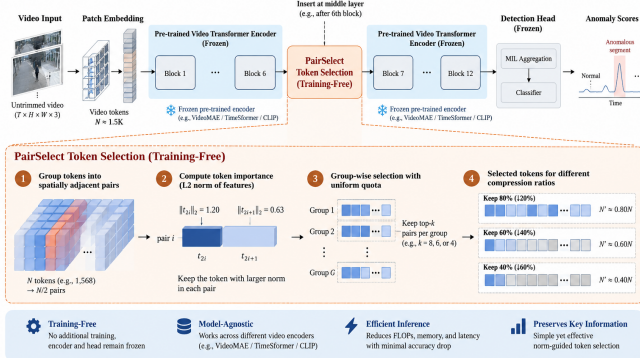


Fig. 1. Overall framework of PairSelect. We insert a training-free token selection module after the middle layer of a pre-trained video transformer encoder. Tokens are grouped into spatially adjacent pairs, and the token with the larger L_2 norm is kept in each pair, followed by a group-wise quota to achieve the desired compression ratio. The selected tokens are fed to the remaining encoder blocks and the frozen detection head to predict video-level anomaly scores.

Table 1. Actual token budgets and analytical reduction of the principal twelve-block computation. Deletion is nominal unless marked actual.

Delete	K	Actual delete	N/K	Block FLOPs ↓
<i>VideoMAE / VideoMAEv2</i>				
20%	1254	20.03%	1.250×	12.05%
40%	941	39.99%	1.666×	23.04%
60%	627	60.01%	2.501×	33.05%
<i>TimeSformer (including CLS)</i>				
20%	1233	21.41%	1.273×	10.94%
40%	897	42.83%	1.749×	21.75%
60%	561	64.24%	2.797×	32.43%

reduction in a frozen MIL pipeline.

3. PAIRSELECT

3.1. Frozen pipeline and design rationale

Let $E_{1:b}$ and $E_{b+1:L}$ denote the prefix and suffix of a pretrained encoder, R its clip readout, and H the temporal detection head. For scheduled clips $\{v_t\}_{t=1}^M$, inference is

$$z_t^{(r)} = R(E_{b+1:L}(S_r(E_{1:b}(v_t)))), \quad \hat{y}^{(r)} = H(\{z_t^{(r)}\}_{t=1}^M). \quad (1)$$

Only S_r , the selection operator at retention budget r , is inserted. We set $L = 12$ and $b = 6$ for all three encoders. The clip count M , output dimension, and head weights remain unchanged. Training-free refers to compression: the original detection head is trained once on dense features for each configuration.

Local redundancy motivates comparing nearby candidates rather than globally ranking all tokens. To avoid an additional mixing operation before an unchanged head, S_r gathers existing vectors. ATS motivates activation magnitude as an inexpensive signal [5], but our block-output L_2 norm is not its attention-weighted value norm. Nor is a high norm an anomaly probability: high-norm transformer tokens can also occur in background regions [17]. We therefore com-

bine magnitude preference with local quotas instead of treating magnitude as a global anomaly oracle.

3.2. Local quotas and pair-member priority

For VideoMAE and VideoMAEv2, the block output is $X \in \mathbb{R}^{N \times D}$ with a known patch grid. We form nonoverlapping horizontal pairs, ordered by native coordinates. Consecutive groups contain at most ten tokens (five pairs). A group with n_g tokens retains

$$q_g(r) = \lfloor rn_g + \frac{1}{2} \rfloor, \quad r \in \{0.8, 0.6, 0.4\}. \quad (2)$$

For pair j with members $(x_{j,0}, x_{j,1})$, define

$$a_j = \arg \max_{u \in \{0,1\}} \|x_{j,u}\|_2, \quad s_j = \max_{u \in \{0,1\}} \|x_{j,u}\|_2. \quad (3)$$

Ties retain native order. Let $w_j = x_{j,a_j}$ and $\ell_j = x_{j,1-a_j}$. If π sorts the m_g pairs by descending s_j , the group’s priority list is

$$P_g = (w_{\pi_1}, \dots, w_{\pi_{m_g}}, \ell_{\pi_1}, \dots, \ell_{\pi_{m_g}}). \quad (4)$$

We take the first q_g entries, then restore the selected indices to native order before gathering. At $r = 0.8$ or 0.6 , every pair in a full group retains a representative; at $r = 0.4$, only four of its five representatives survive. This is a coverage property of the rule, not a guarantee about anomaly preservation. No averaging or rescaling is applied to a selected vector.

3.3. Divided attention and actual budgets

TimeSformer requires a different unit: all T tokens at a spatial location form a trajectory, scored by the maximum member norm. Each group of ten trajectories retains its top q_g trajectories. The current bridge requires the retained spatial count to be a multiple of the native patch width W ; we round the total downward and remove excess quota from tail groups. With $P = 196$, $T = 8$, and $W = 14$,

$$P' = W \left\lfloor \frac{\sum_g q_g(r)}{W} \right\rfloor, \quad K = TP' + 1, \quad (5)$$

where the CLS token is preserved. This width constraint belongs to the current implementation, not to all divided-attention models. Dynamic selections occupy a fixed quota-slot skeleton: gathered content is unchanged, but slot coordinates use approximate anchors rather than an exact reconstruction of the original lattice. Complete trajectories preserve their time indices; the geometry adjustment can empty tail groups.

3.4. Computational accounting

For joint attention with hidden width D and MLP ratio four, the principal block cost in multiply–accumulate operations (MACs) is

$$C_J(N, D) = 12ND^2 + 2N^2D. \quad (6)$$

The linear term includes QKV, output projection, and the MLP; only attention products scale quadratically in N . For standard divided attention, including the temporal projection and replicated CLS in spatial attention [11],

$$C_T(P, T, D) = (17PT + 4T + 8)D^2 + 2D\{PT^2 + T(P + 1)^2\}. \quad (7)$$

Table 2. PairSelect quality at nominal token-deletion budgets. Metrics use a 0–100 scale; Δ_{40} and its paired 95% confidence interval (CI) are percentage-point changes from the corresponding dense run. Dashes mark the two unmeasured budgets, not zero change. Actual token counts are in Table 1.

Encoder	Dense	Delete 20%	Delete 40%	Delete 60%	Δ_{40}	95% CI of Δ_{40}
<i>UCF-Crime — frame ROC-AUC</i>						
VideoMAEv2-B	80.63	—	81.02	—	+0.39	[−0.94, +1.90]
VideoMAE-B	80.55	—	80.39	—	−0.16	[−0.75, +0.50]
TimeSformer-B	78.68	—	78.68	—	−0.001	[−1.12, +1.11]
<i>XD-Violence — frame trapezoidal PR-AUC</i>						
VideoMAEv2-B	76.18	—	76.51	—	+0.32	[−0.06, +0.75]
VideoMAE-B	70.34	—	70.94	—	+0.61	[+0.02, +1.17]
TimeSformer-B	76.69	—	77.37	—	+0.68	[+0.22, +1.13]

The twelve-block cost ratio is $\frac{1}{2}(1 + C_{\text{reduced}}/C_{\text{dense}})$. Table 1 uses $D = 768$ and counts one MAC as two FLOPs. It excludes embedding, readout, normalization, and selection costs. The selector adds norm computation, bounded-size sorting, and gathering; these may have non-negligible runtime despite low arithmetic cost. Token compression, full-encoder FLOPs, latency, and peak memory are therefore distinct quantities.

4. EXPERIMENTS

4.1. Setup and evaluation

We evaluate UCF-Crime [1] with 1,610 training and 290 test videos, and XD-Violence [9] with 3,950 accepted training videos and all 800 test videos. Four of XD’s 3,954 declared training videos are excluded after source-integrity checks. Only visual input and video-level training labels are used. Each of VideoMAEv2-B, VideoMAE-B, and TimeSformer-B is paired with a separately trained UR-DMU head [3]. Head training uses 200 temporal bins, 64 normal and 64 anomalous bags per step, Adam with learning rate 10^{-4} and weight decay 5×10^{-5} , and the final checkpoint after 3,000 steps. This is a controlled dense-feature training protocol, not a reproduction of the detector’s published I3D-feature scores.

The same head checkpoint, preprocessing, and clip schedule are reused across compression settings within a configuration. VideoMAE-family clips yield $8 \times 14 \times 14 = 1568$ tokens; TimeSformer yields the same patch count plus CLS. The completed runs preserve 69,364 UCF and 145,703 XD clips per VideoMAE-family encoder, and 138,853 and 291,781 clips for TimeSformer. Clip totals differ across encoders, but dense and reduced paths match exactly within each configuration. Model weights are frozen during compressed inference.

UCF uses frame ROC-AUC. XD uses trapezoidal PR-AUC under the fixed author-style 16-frame prefix alignment; non-interpolated average precision (AP) is reported separately. These are different integration conventions, not interchangeable names. Differences use 10,000 paired video-level bootstrap resamples stratified by video label. The intervals reflect test-video sampling uncertainty conditional on the reported runs, not variability across training seeds. The 20% and 60% deletion evaluations are a budget-sensitivity extension; no preferred operating point is inferred from missing results.

Table 3. Same-budget references at nominal 40% deletion; entries are Δ from dense in percentage points. Prefix and random use the same group quotas, not additional proposed methods.

Encoder	PairSelect	Prefix	Random
<i>UCF-Crime</i>			
VideoMAEv2-B	+0.39	−0.11	+0.54
VideoMAE-B	−0.16	−0.43	−0.09
TimeSformer-B	−0.001	−0.33	+0.21
<i>XD-Violence</i>			
VideoMAEv2-B	+0.32	−0.41	−0.43
VideoMAE-B	+0.61	+1.14	+1.09
TimeSformer-B	+0.68	+0.10	−0.11

4.2. Direct insertion without head adaptation

Table 2 reports the completed middle-budget results. On UCF, PairSelect changes ROC-AUC by +0.39, −0.16, and approximately 0.00 points for VideoMAEv2, VideoMAE, and TimeSformer, respectively. On XD, the corresponding trapezoidal PR-AUC changes are +0.32, +0.61, and +0.68 points. These point estimates support the feasibility of middle-layer reduction with an unchanged head across the tested encoders. They do not imply a universal improvement: all three UCF intervals include both positive and negative changes.

The alternate XD metric matters. Non-interpolated AP changes from 70.33 to 71.30 for VideoMAEv2, 64.86 to 64.51 for VideoMAE, and 70.86 to 72.04 for TimeSformer. In particular, VideoMAE improves under trapezoidal PR-AUC but decreases under AP. We retain both definitions rather than interpret every positive primary-metric change as an unqualified detection gain.

4.3. Budget behavior and reference selections

The 20%, 40%, and 60% deletion settings test progressively tighter budgets while preserving the same insertion depth. For joint attention, the highest deletion setting begins dropping entire pair representatives. This provides a testable explanation for a possible quality change, not a pre-established breakpoint. For TimeSformer, the actual highest deletion is 64.24%; conclusions must use the executed budget rather than treat its rounding as a measured causal explanation.

Table 3 places the current method alongside fixed-prefix and fixed-seed random selection within each group. Neither is a global uniform sampler, and random masks are reused across videos. Rank-

Table 4. Efficiency ledger for the same dense and PairSelect configurations. F_{blocks} is an analytical principal-block GFLOPs/clip estimate, not a measurement. F_{full} , latency, throughput, and peak allocated/reserved memory require hardware exports and are intentionally unfilled. One MAC is counted as two FLOPs.

Encoder	Delete	Tokens	F_{blocks} GFLOPs	F_{full} GFLOPs	Latency ms/clip	Throughput clips/s	Peak memory alloc./res. GiB
VideoMAEv2-B	Dense	1568	356.99	—	—	—	—/—
	20%	1254	313.99	—	—	—	—/—
	40%	941	274.74	—	—	—	—/—
	60%	627	239.00	—	—	—	—/—
VideoMAE-B	Dense	1568	356.99	—	—	—	—/—
	20%	1254	313.99	—	—	—	—/—
	40%	941	274.74	—	—	—	—/—
	60%	627	239.00	—	—	—	—/—
TimeSformer-B	Dense	1569	389.81	—	—	—	—/—
	20%	1233	347.15	—	—	—	—/—
	40%	897	305.01	—	—	—	—/—
	60%	561	263.40	—	—	—	—/—

ings vary: random is higher on all three UCF configurations, whereas PairSelect is higher on two XD configurations. Thus these references do not establish universal superiority of the norm signal. Their static native-position gathering also differs from the dynamic anchor-slot assignment, so this is a matched-budget reference rather than an isolated norm-only ablation.

4.4. FLOPs and hardware efficiency

Table 4 separates analytical savings from the hardware measurements still required. At the middle budget, principal-block FLOPs decrease by 23.04% for joint attention and 21.75% for divided attention. At the highest deletion setting, the corresponding analytical reductions are 33.05% and 32.43%. Because the first six blocks remain dense, these savings are smaller than the token-removal percentages.

The measurement protocol fixes the GPU, input shape, batch size, numerical precision, and attention backend across dense and reduced runs. Full-encoder timing includes embedding, all blocks, the selector, index gathering, and readout. Pure diagnostic CPU exports are disabled consistently, but actual selection work is not removed. The planned measurements use 20 warm-up passes, synchronized CUDA-event timing, and three independent process repeats on the same representative non-test clips. Throughput counts completed clips at the stated batch; single-clip latency is reported at batch one. Peak allocated and reserved memory are recorded after synchronized reset of peak counters. Hardware and software identifiers must accompany the completed table.

A separate synchronized wall-clock measurement is required for decoding, host-device transfer, feature writing, and temporal-head inference. Since the encoders use different clip schedules, clips/s is not an end-to-end cross-encoder speed ranking. Nor does suffix token reduction ensure lower whole-encoder peak memory: the dense prefix can retain the peak, and selector temporaries can add overhead. No measured speedup or memory reduction is claimed from the present analytical table.

5. SCOPE AND DISCUSSION

The evidence covers three backbones from two principal attention families and one downstream detector family. A shared selection

rule does not imply identical behavior across architectures, datasets, or metrics. Mid-depth insertion and the group size are fixed design choices, not proven optima. The current controls do not substitute for a direct comparison with a standard external reducer such as ATS or ToMe. Earlier engineering diagnostics accessed UCF test data; the study is therefore not presented as wholly test-blind development. Additional budgets must be reported together rather than selected post hoc to imply an independently validated optimum.

The local-coverage rule constrains where token removal occurs, but it does not identify abnormal locations or guarantee preservation of short events. Dynamic anchor positions are an implementation approximation. The method reduces internal activation processing, not model weights, output feature dimension, or the number of detector inputs. Its deployment benefit ultimately depends on measured arithmetic cost, execution overhead, and hardware behavior together.

6. CONCLUSION

PairSelect reduces internal video-token computation without reducing the existing clip schedule or retraining the anomaly detection pipeline. Local quotas, magnitude-guided representatives, and non-mixing gathers provide a configurable selection rule for joint and divided attention. The completed middle-budget results show small primary-metric point-estimate changes across three encoders and two datasets, alongside a 21.75–23.04% analytical reduction in principal-block operations. Completing the budget ladder and hardware measurements will determine which operating points offer a useful quality–efficiency trade-off in practice.

7. COMPLIANCE WITH ETHICAL STANDARDS

This work uses the existing UCF-Crime and XD-Violence video benchmarks; it does not report newly collected participant data. **Author confirmation required:** document the applicable dataset permissions and institutional determination of whether ethics approval or exemption is needed. No approval or waiver is asserted in this working draft.

8. ACKNOWLEDGMENTS

Author confirmation required: insert the funding and conflict-of-interest declarations, including an explicit absence-of-conflict statement when applicable. No funding or conflict declaration is inferred here.

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